Bayesian Modelling: Monte Carlo Sampling and Integration

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[0] C. P. Robert & G. Casella. Monte Carlo Statistical Methods. 2nd edition, Springer, 2004.

Overview

Inversion sampling

Transformations

Rejection sampling

Importance sampling

Self-normalised importance sampling

Introduction

- Most Bayesian posteriors are not conjugate, and have no closed-form representation.
- Often, the only way to examine them is to draw samples from the posterior.
- Thus, generation of random numbers is a key tool in Bayesian modelling.
- Random numbers are also a key numerical method for approximating integrals, e.g. for means, credible sets, etc.

Inversion sampling

Inversion sampling

- If $F(x)$ is a continuous CDF and $U \sim U(0,1)$, then $F^{-1}(U) \sim F$:

$$\mathbb{P}(F^{-1}(U) \leq x) = \mathbb{P}(U \leq F(x)) = F(x).$$

- To sample from a density $f(x)$, compute its inverse CDF, $F^{-1}(x)$, sample $U \sim U(0,1)$, and return $F^{-1}(U)$.
- Example: $f(x) = \lambda e^{-\lambda x}$ yields

$$F^{-1}(u) = \frac{-\log(1-u)}{\lambda}.$$

- Problems when F^{-1} is hard to evaluate, or in dimension > 1.

Transformations

Other transformations

- Sometimes random variables are transformations of simpler ones.
- E.g. if $X \sim N(0,1)$ then $X^2 \sim \chi^2_1$.
- If $X \sim N(0,1)$ and $Y \sim \chi^2_n$ are independent, then

$$\frac{Z}{\sqrt{V/n}} \sim t_n.$$

- If $(X_1, \ldots, X_n) \overset{i.i.d.}{\sim} \text{Exp}(\lambda)$, then

$$\sum_{j=1}^{n} X_i \sim \Gamma(n, \lambda).$$

- If $X \sim \Gamma(r, \lambda)$ and $Y \sim \Gamma(s, \lambda)$ are independent, then

$$\frac{X}{X+Y} \sim \text{Beta}(r, s).$$

The Box–Muller algorithm

- Let $U_1, U_2 \sim U(0,1)$ be independent.
- Define

$$R := \sqrt{-2\log(U_1)},$$
$$\theta := 2\pi U_2.$$

Then

$$X := R\cos(\theta),$$
$$Y := R\sin(\theta),$$

are i.i.d. and $N(0,1)$-distributed.

Rejection sampling

Rejection sampling

- Suppose density $f(x)$ is difficult to sample, e.g. no closed form inverse CDF and no known transformation.
- Idea: sample from an easier density $g(x)$, and either accept or reject each draw.
- Accepted draws will come from density $f(x)$.

Rejection sampling algorithm

- Assume $Y \sim g$ and there exists $c < \infty$ such that whenever $f(y) > 0$ we have
$$\frac{f(y)}{g(y)} \leq c.$$
- Generate $Y \sim g$.
- Generate $U \sim U(0, 1)$.
- Accept Y if
$$U \leq \frac{f(Y)}{cg(Y)},$$
else reject. Note that the right-hand side is ≤ 1 by construction.

Rejection sampling: analysis

- We need an average of $1/c$ proposals for one acceptance:

$$\begin{aligned}\mathbb{P}(\text{accept}) = \mathbb{E}[\mathbb{P}(\text{accept}|Y)] &= \frac{1}{c}\mathbb{E}\Big[\frac{f(Y)}{g(Y)}\Big] \\ &= \frac{1}{c}\int \frac{f(y)}{g(y)} g(y)\mathrm{d}y = \frac{1}{c}.\end{aligned}$$

- Accepted draws have density f by Bayes' rule:

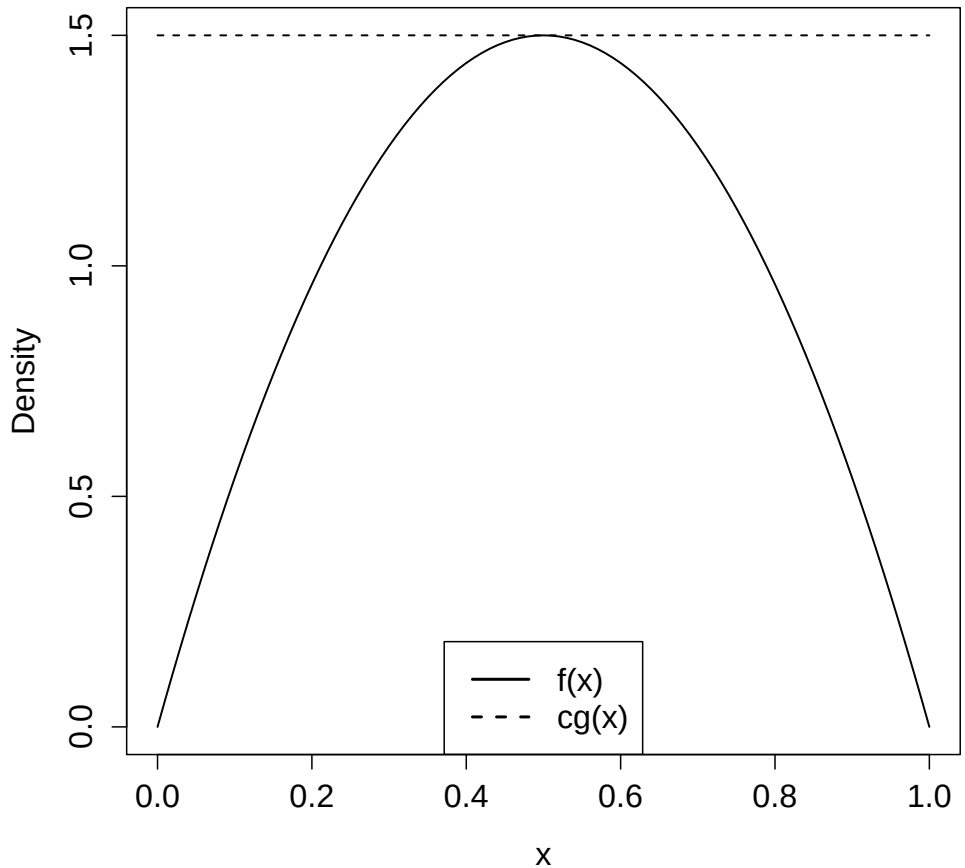
$$\begin{aligned}\mathbb{P}(Y \in \mathrm{d}y|\text{accept}) &= \frac{\mathbb{P}(\text{accept}|Y=y)g(y)}{\mathbb{P}(\text{accept})}\mathrm{d}y \\ &= \frac{f(y)}{cg(y)}\frac{g(y)}{1/c} = f(y)\mathrm{d}y.\end{aligned}$$

Rejection sampling: example

- Target: $f(x) = 6x(1-x)$ for $x \in [0,1]$, i.e. $X \sim \text{Beta}(2,2)$.
- Proposal: $Y \sim U(0,1)$.
- We have

$$\frac{f(x)}{g(x)} = 6x(1-x) \leq 1.5.$$

Rejection sampling: example



- Proposals are sampled uniformly under the graph of $cg(x)$. Those which fall under the graph of $f(x)$ are accepted.

Importance sampling

Monte Carlo integration

- Nearly all statistical quantities can be cast as integrals,

$$I := \int_{\Omega} f(x)\pi(x)\mathrm{d}x,$$

  where $f : \Omega \to \mathbb{R}$ is a function and $\pi(x)$ is a pdf (or pmf).
- E.g. $f(x) = x$ and $\pi(x) = p(x|y)$ is the posterior mean, $f(x) = \mathbb{1}_A(x)$ is the probability of event A, etc.
- By the law of large numbers, if $(X_1, \ldots, X_N) \overset{i.i.d.}{\sim} \pi$, then

$$\widehat{I}_N := \frac{1}{N}\sum_{i=1}^{N} f(X_i) \to I$$

  almost surely as $N \to \infty$, and $\widehat{I}_N$ is unbiased:

$$\mathbb{E}[\widehat{I}_N] = \mathbb{E}[f(X)] = I.$$

Importance sampling

- Variance is a natural measure of how well $\widehat{I}_N$ approximates I:

$$\mathrm{Var}(\widehat{I}_N) = \frac{1}{N}\mathrm{Var}(f(X)).$$

- Let $\phi(x)$ be another pdf with $\phi(x) > 0$ whenever $\pi(x) > 0$, known as the proposal distribution.
- Consider $(X_1, \ldots, X_N) \overset{i.i.d.}{\sim} \phi$ and the importance sampling (IS) estimator

$$\widehat{I}_N^{\phi} := \frac{1}{N}\sum_{i=1}^{N} f(X_i)\frac{\pi(X_i)}{\phi(X_i)}.$$

Importance sampling: performance

- Th IS estimator $\widehat{I}_N^{\phi}$ is also unbiased:

$$\mathbb{E}_{\phi}[\widehat{I}_N^{\phi}] = \mathbb{E}_{\phi}\left[f(X)\frac{\pi(X)}{\phi(X)}\right] = \int_{\Omega} f(x)\frac{\pi(x)}{\phi(x)}\phi(x)\mathrm{d}x = \mathbb{E}_{\pi}[f(X)],$$

  where the subscript on $\mathbb{E}$ emphasizes that $X \sim \phi$.
- IS can be preferred if ϕ is easier to sample from than π.
- It can also help reduce variance:

$$\begin{aligned}\mathrm{Var}(\widehat{I}_N^{\phi}) &= \frac{1}{N}\mathrm{Var}_{\phi}\left(f(X)\frac{\pi(X)}{\phi(X)}\right) \\ &= \frac{1}{N}\left(\mathbb{E}_{\pi}\left[f(X)^2\frac{\pi(X)}{\phi(X)}\right] - \mathbb{E}_{\pi}[f(X)]^2\right).\end{aligned}$$

Optimal importance sampling

- Choosing $\phi(x) = f(x)\pi(x)/\mathbb{E}_\pi[f(X)]$ yields

$$\mathrm{Var}(\widehat{I}_N^\phi) = \frac{1}{N}(\mathbb{E}_\pi[f(X)\mathbb{E}_\pi[f(X)]] - \mathbb{E}_\pi[f(X)]^2) = 0.$$

- But the denominator of ϕ is the integral to approximate in the first place! So the resulting algorithm isn't implementable.
- Useful heuristic: design $\phi(x) \propto f(x)\pi(x)$, approximately.
- Unlike rejection sampling, $\phi(x)/\pi(x)$ need not be bounded. Though for good performance we do need

$$\mathbb{E}_\phi\left[\left(\frac{\pi(X)}{\phi(X)}\right)^2\right] = \mathbb{E}_\pi\left[\frac{\pi(X)}{\phi(X)}\right] < \infty,$$

  i.e. ϕ should have heavier tails than π.

Example: computing a tail probability

- Suppose

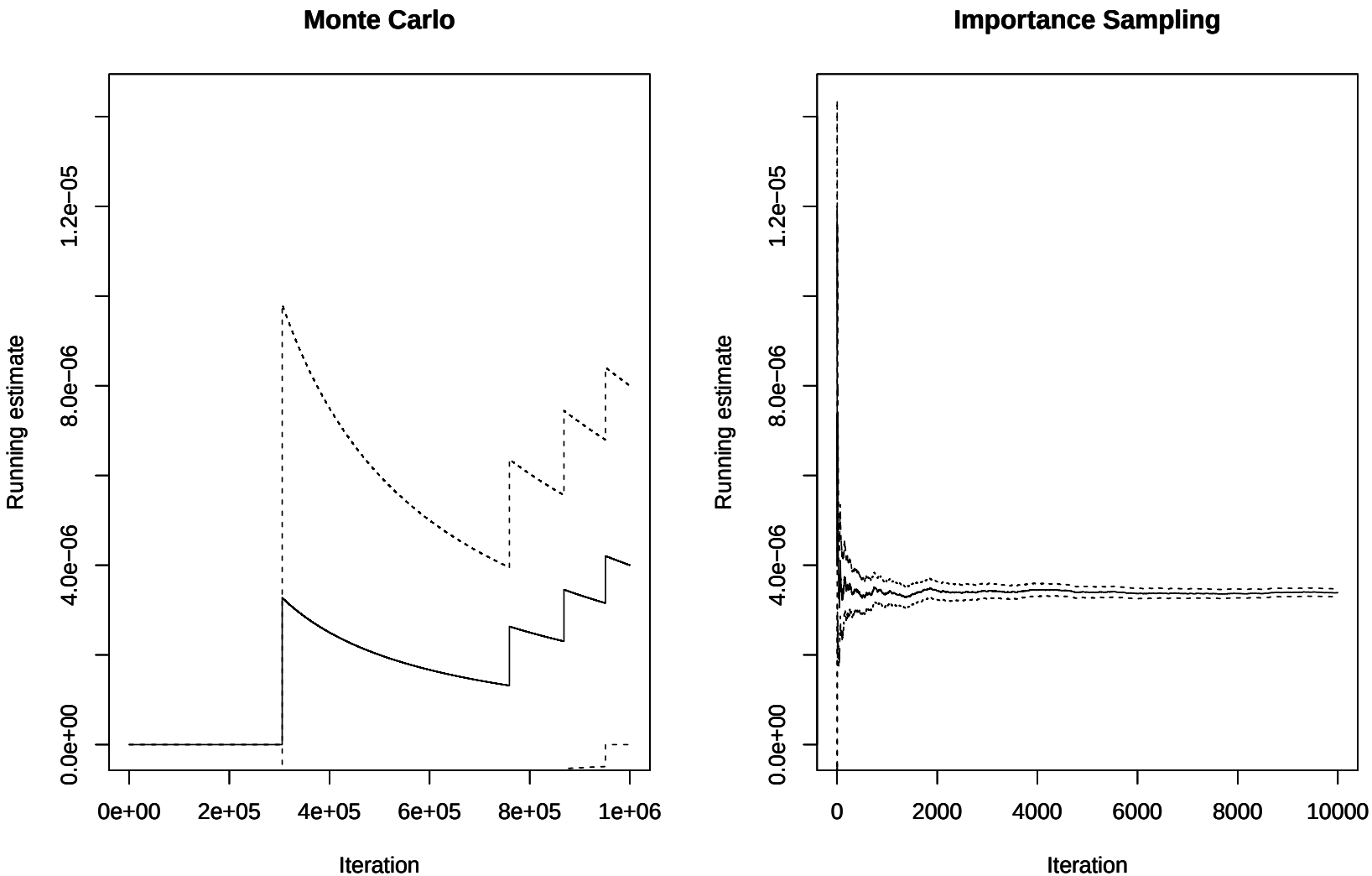
$$f(x) = \mathbb{1}_{[4.5,\infty)}(x),$$
$$\pi(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2},$$
$$\Omega = \mathbb{R}.$$

- We choose ϕ to be the shifted exponential density,

$$\phi(x) = e^{-(x-4.5)},$$

  for $x \geq 4.5$.
- This ϕ is easy to sample in e.g. `R` via `4.5 + rexp(1)`.
- True probability $\approx 3.4 \times 10^{-6}$.

Example: computing a tail probability



Self-normalised importance sampling

Importance sampling for Bayesian posteriors

- Consider importance sampling for the Bayesian posterior

$$\pi(x) = p(x|y) = \frac{p(y|x)p(x)}{\int_\Omega p(y|x)p(x)\mathrm{d}x} =: \frac{p(y|x)p(x)}{Z},$$

  where $x \in \Omega$ is the parameter of interest, y is data, $p(x)$ is the prior, and $p(y|x)$ is the likelihood.
- Unless the prior and likelihood are conjugate, the normalising constant Z is as hard to evaluate as an integral of interest.
- If we can't compute Z, we (usually) can't sample from $p(x|y)$ and can't evaluate the importance weight $\pi(x)/\phi(x)$ for IS.

Self-normalised importance sampling

- Consider an unnormalised target distribution

$$\pi(x) = \frac{\eta(x)}{Z},$$

where $\eta(x)$ can be evaluated but Z cannot.

- For $(X_1, \ldots, X_N) \overset{i.i.d.}{\sim} \phi$, define the unnormalised importance weights and the self-normalised IS estimator:

$$w(X_i) := \frac{\eta(X_i)}{\phi(X_i)},$$

$$\widehat{I}_N^{\phi,\mathrm{SN}} := \sum_{i=1}^{N} \frac{w(X_i)}{\sum_{j=1}^{N} w(X_j)} f(X_i).$$

Self-normalised importance sampling

$$\widehat{I}_N^{\phi,\mathrm{SN}} := \sum_{i=1}^{N} \frac{w(X_i)}{\sum_{j=1}^{N} w(X_j)} f(X_i).$$

- Notice that Z cancels!
- But $\widehat{I}_N^{\phi,\mathrm{SN}}$ is biased.
- The LLN does not hold because the summands are correlated by self-normalisation.
- We need new theoretical tools.

Slutsky's theorem

- Let $(X_n)_{n\geq 1}, (Y_n)_{n\geq 1}$ be sequences of random variables.
- Suppose $X_n \xrightarrow{\mathrm{d}} X$ as $n \to \infty$, where X is a random variable, and
- $Y_n \xrightarrow{\mathbb{P}} c$ as $n \to \infty$, where $c \neq 0$ is a constant.
- Then, as $n \to \infty$,

$$X_n + Y_n \xrightarrow{\mathrm{d}} X + c,$$

$$X_n \cdot Y_n \xrightarrow{\mathrm{d}} X \cdot c,$$

$$X_n / Y_n \xrightarrow{\mathrm{d}} X / c.$$

- Proof: the hypotheses imply $(X_n, Y_n) \xrightarrow{\mathrm{d}} (X, c)$. The conclusion follows from the continuous mapping theorem.

Consistency of self-normalised IS

$$\sqrt{N}(\widehat{I}_N^{\phi,\mathrm{SN}} - \mathbb{E}_\pi[f(X)]) = \frac{\sqrt{N}\sum_{i=1}^N \frac{w(X_i)}{NZ}(f(X_i) - \mathbb{E}_\pi[f(X)])}{\sum_{j=1}^N \frac{w(X_j)}{NZ}}$$
$$=: \frac{A_N}{B_N}.$$

- By the central limit theorem, as $N \to \infty$,

$$A_N \xrightarrow{\mathrm{d}} N\left(0, \mathbb{E}_\pi\left[\frac{\pi(X)}{\phi(X)}\{f(X) - \mathbb{E}_\pi[f(X)]\}^2\right]\right).$$

- By the law of large numbers, as $N \to \infty$,

$$B_N \xrightarrow{\text{a.s.}} \mathbb{E}_\phi\left[\frac{\eta(X)}{Z\phi(X)}\right] = \mathbb{E}_\pi[1] = 1.$$

Consistency of self-normalised IS

- By Slutsky's theorem

$$\sqrt{N}\widehat{I}_N^{\phi,\mathrm{SN}} \to N\left(\mathbb{E}_\pi[f(X)], \mathbb{E}_\pi\left[\frac{\pi(X)}{\phi(X)}\{f(X) - \mathbb{E}_\pi[f(X)]\}^2\right]\right).$$

- The same sample $(X_1, \ldots, X_N) \overset{i.i.d.}{\sim} \phi$ can be used to estimate the asymptotic variance:

$$\begin{aligned}
&\sum_{i=1}^{N} \frac{w(X_i)^2 (f(X_i) - \widehat{I}_N^{\phi,\mathrm{SN}})^2}{\left(\sum_{j=1}^{N} w(X_j)\right)^2} \\
&\approx \frac{1}{N}\mathbb{E}_\pi\left[\frac{\pi(X)}{\phi(X)}(f(X) - \mathbb{E}_\pi[f(X)])^2\right].
\end{aligned}$$

- Even when Z is available, $\widehat{I}_N^{\phi,\mathrm{SN}}$ can have smaller variance than $\widehat{I}_N^{\phi}$, but is biased.

Example: non-conjugate posterior

- You're asked to evaluate the efficacy of a new cancer treatment based on whether it shrinks tumours in observational pilot studies, prior to a potential clinical trial.
- The pilot studies tracked n patients who tried the treatment, among whom x saw their tumour shrink.
- A binomial model is natural: $X|\pi \sim \text{Bin}(n, \pi)$.
- An expert oncologist suggests a Gaussian prior for π conditioned to lie in $[0, 1]$:

$$p(\pi) \propto \frac{\mathbb{1}_{[0,1]}(\pi)}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{-(\pi-\mu)^2}{2\sigma^2}\right).$$

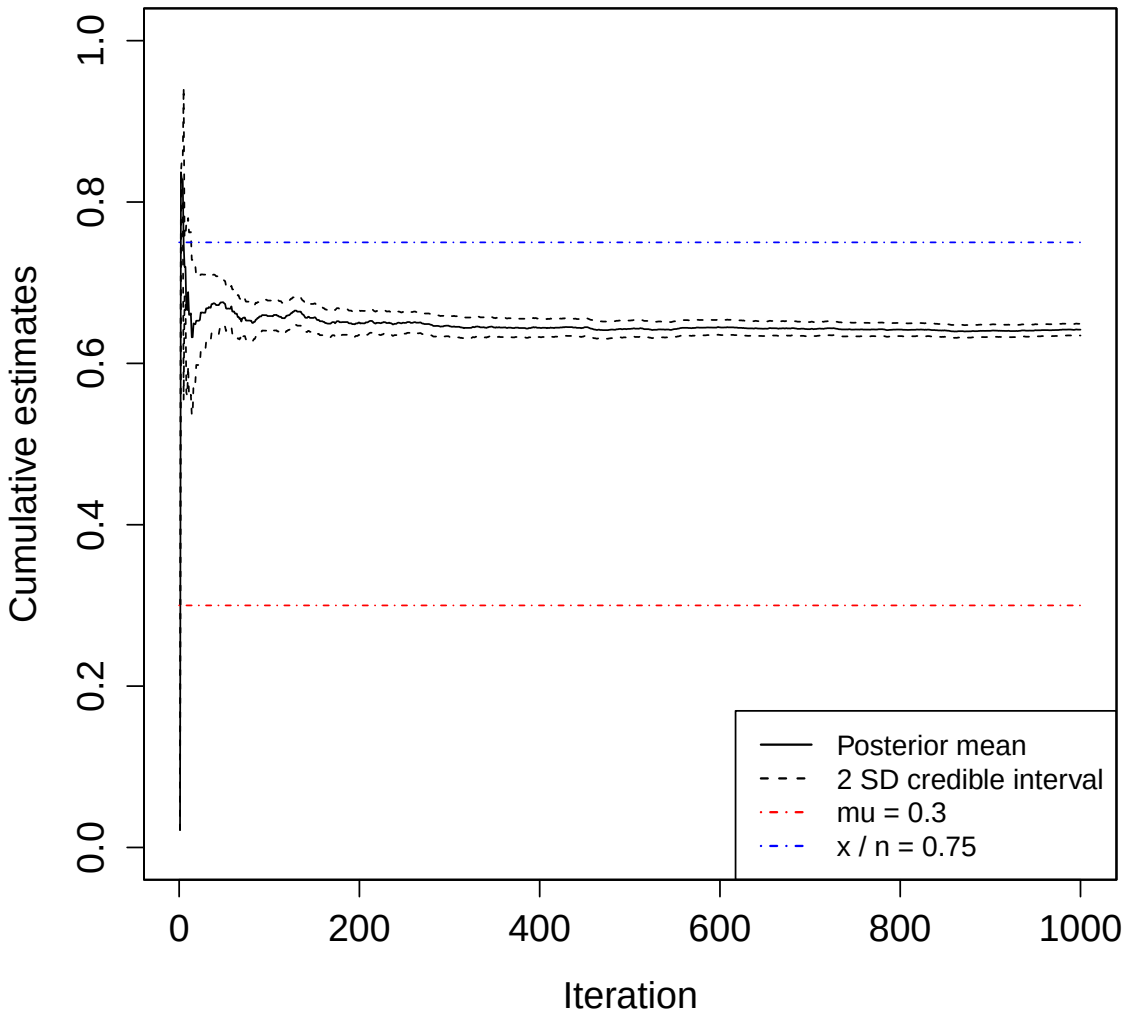
- This results in the posterior

$$p(\pi|x) \propto \mathbb{1}_{[0,1]}(\pi)\pi^x(1-\pi)^{n-x} \exp\left(\frac{-(\pi-\mu)^2}{2\sigma^2}\right).$$

Example: non-conjugate posterior

- For sake of illustration set $n = 20$, $x = 15$, $\mu = 0.3$, $\sigma = 0.2$.
- The posterior is intractable: we cannot plot normalised densities etc. without resorting to numerical integration or Monte Carlo.
- We approximate the posterior mean using SNIS with uniform proposal $\phi(x) = 1$ on $[0, 1]$.

Example: non-conjugate posterior mean



Example: non-conjugate posterior density

